

Paul Leger: Report from 2025

Associate Professor & Software Engineer
Civil Engineering School
Catholic University of the North
Larrondo 1281, Coquimbo, Chile

Email: pleger@ucn.cl
Homepage: <http://pleger.cl>
Pragmatics lab: <http://pragmatics-lab.com> (Director)
LinkedIn: <http://cl.linkedin.com/in/plegerm>

Personal

Name: Paul Leger
Hometown: Coquimbo, Chile
Nationality: Chilean



Publications

My research profile IDs are *ORCID* (0000-0003-0969-5139), *Web of Science* (Q-6174-2017), and *Scopus* (36087550600).

NOTE: For conferences/workshops, I use the CORE ranking, which belongs COMputing REsearch and Education Association of Austral-asia (near to the publication year). CORE is available on <http://portal.core.edu.au/conf-ranks>.

Journals (Indexed)

- Canquil, Diego, Daniel SanMartín, Irene Inoquio-Renteria, and Paul Leger (Jan. 2026). "A Systematic Literature Review on Machine Learning Techniques for Predicting Household Water Consumption". In: *Water Resources Management* 40.2, p. 83. doi: 10.1007/s11269-026-04496-4. **JCR-WOS**.
- Leger, Paul, David Moreno-Lumbreras, Felipe Ruiz, Nicolás Sepúlveda, Ismael Figueroa, and Nicolás Cardozo (Feb. 2026). "The Stackage Repository: An Exploratory Study of its Evolution". In: *Journal of Computer Languages* 86, p. 101386. doi: 10.1016/j.cola.2026.101386. **JCR-WOS**.
- Leger, Paul, Daniel San-Martín, Diego Nuñez, and Tomás Vélez (Feb. 2026). "An Agent-Based Model Reflective Architecture for Social Network Sites". In: *Science of Computer Programming* 252, p. 103452. doi: 10.1016/j.scico.2026.103452. **JCR-WOS**.
- Abdi, Farshid, Shaghayegh Abolmakarem, Amir Karbassi-Yazdi, Paul Leger, Yong Tan, and Giuliani Coluccio (Jan. 2025). "Predicting Patients' Revisit Intention Based on Satisfaction Scores: Combination of Penalized Regression and Neural Networks". In: *IEEE ACCESS* 13, pp. 2783–2800. doi: 10.1109/ACCESS.2024.3522767. **JCR-WOS**.
- Karbassi, Amir, Yang Tan, and Paul Leger (Jan. 2025). "Clustering Financial Institutions in Countries Based on a Hybrid Random Forest and Induced Ordered Weighted Averaging". In: *Journal of Management Analytics*, pp. 1–30. doi: 10.1080/23270012.2025.2454674. **JCR-WOS**.
- Terán, Oswaldo, Paul Leger, and Manuela López (Apr. 2025). "Factors that Drive Market Share and the Oligopolistic Character of Cross-border B2C Ecommerce: An agent-based scenario analysis approach". In: *Simulation: Transactions of the Society for Modeling and Simulation International* 0.0, pp. 1–24. doi: 10.1177/00375497241296542. **JCR-WOS**.

Conferences (SCOPUS)

- Liu, Shijin, Hiroaki Fukuda, and Paul Leger (Mar. 2026). "SOM-based Continuous XGBoost Approach for DDoS Attack Detection and Mitigation in Edge SDN". In: *Workshop on Programmable and Learning Architectures for Intelligent Networks*. Athens, Greece. doi: 10.1109/ICIN69025.2026.11481839. **SCOPUS**. **Rank National-France (CORE ranking). To appear.**

SanMartin, Daniel, Walter Carmago, and Paul Leger (Sept. 2026). "Towards a Constraint-Based Approach for Quantum Circuit Modeling". In: *III Taller Latinoamericano en Ingeniería y Software Cuántico*. Mexico City, Mexico. **SCOPUS. Rank C (CORE ranking). To appear.**

Ogino, Toshihiro, Hiroaki Fukuda, and Paul Leger (July 2025). "Stepwise Distributed Execution for Load Reduction in Machine Learning Systems". In: *International Symposium on Parallel and Distributed Computing (ISPDC)*. Rennes, France, pp. 127–131. DOI: 10.1109/ISPDC67428.2025.00028. **SCOPUS. Rank C (CORE ranking).**

You, Koshi, Hiroaki Fukuda, and Paul Leger (Apr. 2025). "The Impact of Context-Oriented Programming on Declarative UI Design in React". In: *Evaluation of Novel Approaches to Software Engineering (ENASE)*. Porto, Portugal, pp. 700–707. DOI: 10.5220/0013432300003928. **SCOPUS. Rank B (CORE ranking).**

Workshops (SCOPUS)

Montes-León, Sergio R., Paul Leger, David Moreno-Lumbreras, Daniel San Martín Santibáñez, and Antonio J. Romero-Barrera (Sept. 2026). "Are Software Visualization Tools Actually Usable, or Just a Graveyard of Good Ideas? A Systematic Review and Replicability Assessment". In: *IEEE Working Conference on Software Visualization (VISOFT)*. Benevento, Italy. **SCOPUS. Rank B(CORE ranking). To appear.**

Congresses (Conferences/Workshops not indexed)

Note: These papers are just presentations of technical reports or working papers.

Moreno-Lumbreras, David, Paul Leger, Sergio Montes-León, Jesus Gonzalez-Barahona, and Gregorio Robles (Sept. 2025). "Evaluating the Comprehension of the Stackage Ecosystem: A Comparison between VR and 2D Visualizations". In: *Registered Report: 2025 IEEE International Conference on Software Maintenance and Evolution (ICSME)*. Auckland, New Zealand. DOI: 10.2139/ssrn.5362665. **Registered Report.**

Posters

Scientific Services

Manager (x 2)

- **Director on the Engineering School at UCN**, Coquimbo (Chile), 2023 - present, <https://www.ucn.cl>.
- **Pragmatics Lab. Founder and director of the research laboratory in software development and simulations of agent-based models**, Virtual (Chile), 2018 - present, <https://pragmatics-lab.com>.

Journal Editor (x 3)

- **Associate Editor on Frontiers in Computer Science**, ISSN: 2624-9898, Frontiers, 2025 - present, JCR-WOS.
- **Editorial Member on Journal of Innovations in Digital Marketing**, ISSN: 2765-8341, Luminous Insights, 2021 - present.
- **Guest Editor on Frontiers in Computer Science - Special Issue on Artificial Intelligence for Software Engineering: Advances, Applications, and Implications**, ISSN: 2624-9898 - Frontiers, 2025, NOTE: 2025.

Journal Reviewer (x 2)

- **Scientific Data**, ISSN: 2052-4463, Nature, 2026, JCR-WOS.
- **Journal of Systems and Software**, ISSN: 0164-1212, ELSEVIER, 2025, JCR-WOS.

Program Committee Member (x 6)

- **Chilean Workshop on Quantum Computing and Quantum Software Engineering (QSQCE-Chile)**, Chile, 2026, SCOPUS, <https://jcc2026.utalca.cl/wsqsqce-en.html>.
- **International Conference on Software Technologies (ICSOFT)**, Porto (Portugal), 2026, SCOPUS, <https://icsoft.scitevents.org/ProgramCommittee.aspx>.
- **(SPLASH) 2025 Workshop on Virtual Machines and Language Implementations**, Singapore, 2025, SCOPUS, <https://conf.researchr.org/home/icfp-splash-2025/vmil-2025>.
- **International Conference on Software Maintenance and Evolution (ICSME) - Industry Track**, Auckland (New Zealand), 2025, SCOPUS, <https://conf.researchr.org/track/icsme-2025/icsme-2025-industry-track>.
- **XXXVI International Marketing Congress (AEMARK)**, Murcia (Spain), 2025, <https://aemarkcongresos.com/areas-tematicas-y-comite-cientifico>.
- **International Conference on Software Technologies (ICSOFT)**, Bilbao (Spain), 2025, SCOPUS, <https://icsoft.scitevents.org/ProgramCommittee.aspx>.

Evaluator (x 7)

- **Evaluation committee of FONDEF-IT (Health-IT)**, ANID (Chile), 2026, <https://anid.cl/concursos/concurso-de-investigacion-tecnologica-idea-2026/>.
- **President of the Ph.D thesis commission: "Diversidad de Licencias en Software Libre"**, Universidad Rey Juan Carlos (Spain), 2025, <https://en.urjc.es>.
- **President of the Ph.D thesis commission: "Integrated Mathematical Models for Agricultural Decision-Making Under Uncertainty: Efficiency; Water Optimization and Resilience in Olive Production."**, Universidad Católica del Norte (Chile), 2025, <https://www.ucn.cl>.
- **Evaluator of Fondef-IDeA Projects (x 4)**, ANID (Chile), 2023 - 2025, <https://anid.cl/concursos/concurso-idea-id-tecnologias-avanzadas-2025/2025>.

Software Development for Research

My software projects are more related to creating complex software artifacts than developing traditional industrial software (e.g., Inventory Management). Source: <http://pleger.cl/software>.

PL/SE research with AI: Implementation with many upgrades of my previous research in Programming Language and Software Engineering with Artificial Intelligence agents (Codex): AspectScript, Expressive Stateful Aspect (ESA-myPhD), DeloreanGo (adaptation from DeloreanJs), Sync/CC, and EMA-Kotlin (adaptation from EMA-js). <https://pleger.github.io/PL-SE-research-with-AI>, 2026. [ARTIFICIAL INTELLIGENCE].

Academic Experience

Undergraduate (x 12)

- Complex Engineering Problems (x 4)**, Information and Information Technology, Catholic University of the North, 2024 - 2026.
- Programming Languages (x 8)**, Computer Civil Engineering, Catholic University of the North, 2021 - 2025.

Thesis Advising (x 27)

- Postgraduate (x 3)**, Ph. D and Master, Catholic University of the North, 2017 - 2025.
- Undergrad (x 24)**, Engineering, Catholic University of the North, 2013 - 2025.